Experiment 2: Prereading Questions

Each participant gets four prequestions and eight final questions (four prequestions and four new questions). Each question serves as a prequestion and a new question and is counterbalanced across participants. That is, Participant X gets prequestions 1-5 and receives questions 6-10 as new questions while Participant Y gets prequestions 6-10 and receives questions 1-5 as news questions. The control group only receives the final test questions. Each participant is presented all the questions for the final test.

1. What percent of the global budget does the Gates Foundation devote toward the eradication of polio? (1 point)
2. Is there a cure for polio? (1 point)
3. In which countries is polio still a major concern? (4 points)
4. What measures are included in the modern treatment of polio?" (4 points)
5. Since the development of the polio vaccine, the global number of polio cases per year has dropped from _____ to _____. (2 points)

6. What are the symptoms of tuberculosis? (4 points)
7. What percentage of new infections will turn into active cases of tuberculosis? (1 point)
8. What is the treatment for tuberculosis? (3 points)
9. Can people with latent tuberculosis spread the disease? (1 point)
10. How is tuberculosis spread through the air? (3 points)

Yellow = percent

Red = yes/no

Green = multi-part

Gates Foundation text. Targeted sentences are underlined:

The Bill & Melinda Gates Foundation has quickly become a major influence upon global health. The approximately $800 million that the foundation gives every year for global health initiatives approaches the annual budget of the United Nations World Health Organization. The Gates Foundation has been particularly active in promoting the development and administration of vaccines. **The foundation currently provides 17% of the global budget for the attempted eradication of polio**. The Gates Foundation has also been a strong supporter of new tuberculosis vaccine development. These efforts demonstrate the role that philanthropy can play in eliminating disease.

Diseases like polio and tuberculosis have affected man since prehistory. Polio, sometimes called infantile paralysis, is an acute viral infectious disease that spreads easily through human-to-human contact. Although around 90% of polio infections cause no symptoms at all, in a small minority of cases the virus enters the central nervous system and causes acute paralysis. Tuberculosis (TB) is an often deadly infectious disease caused by mycobacteria, mainly Mycobacterium tuberculosis. **The classic symptoms of TB are a chronic cough with blood-tinged sputum, fever, night sweats, and weight loss.**

In endemic areas, wild polioviruses can infect virtually the entire human population. **Tuberculosis is spread through the air when people who have the disease cough, sneeze, or spit.** One third of the world's current population have been infected with M. tuberculosis. **New infections occur at a rate of one per second, but only 10% of these cases will develop into active TB.** The Gates Foundation and other non-governmental organizations recognize that vaccination holds the key to preventing the rapid transmission of infectious diseases. Once a vast majority of people are vaccinated, it is much more difficult for an outbreak of a disease to occur or for the disease to spread. This effect is called "herd immunity." **Widespread vaccination led to the eradication of polio in the Americas and Europe, but polio is still endemic to Nigeria, India, Pakistan, and Afghanistan.**

The "herd immunity" provided by vaccination is less costly and more effective at stopping the spread of disease than conventional measures like quarantine. **Only people with active, not latent, TB can transmit the disease.** The chain of transmission for TB can be broken by isolating patients with active disease and starting effective antibiotic treatment. Historically, quarantine has been less successful in preventing the spread of polio.

The Gates Foundation's focus on vaccination is also critical since many diseases cannot be successfully treated by conventional antibiotics. **Treatment for active TB requires a long-term course of two antibiotics to kill the bacteria.** Antibiotic resistance is a growing problem in (extensively) multi-drug-resistant tuberculosis. **Polio has no cure.** The focus of modern treatment has been on providing relief of symptoms and preventing complications. **Supportive measures include antibiotics to prevent infections in weakened muscles, analgesics for pain, moderate exercise, and a nutritious diet.**

When effective vaccines are available, non-governmental organizations should focus on increasing rates of vaccination. **Thanks to the polio vaccines developed by Jonas Salk in 1952 and Albert Sabin in 1962, the global number of polio cases per year has dropped from many hundreds of thousands to around a thousand.** Many countries use the Bacillus Calmette–Guérin (BCG) vaccine as part of their TB control programs, especially for infants. Unfortunately, BCG does not reliably protect adults. For this reason, the Gates Foundation recently announced a $200 million grant to the Aeras Global TB Vaccine Foundation for clinical trials on up to six different TB vaccine candidates currently in the pipeline. Since the oral polio vaccine is both highly effective and cheap (about $0.10), polio eradication efforts focus on basic health infrastructure and vaccine distribution.

The scale of the Gates Foundation and the way it seeks to apply business techniques to giving allows it to play a powerful role in the global health movement. This new approach to philanthropy has been dubbed "philanthrocapitalism." "Philanthrocapitalism," or venture philanthropy, takes concepts and techniques from venture capital finance and high technology business management and applies them to problems like the spread of disease in developing countries. Hopefully, funding from organizations like the Gates Foundation will lead to the eradication of diseases like polio and TB

Eye measures: **FIXATION Ratios and Durations:**

Total Duration - The total duration of all fixations in a target region.

regressions to sentences

Re-reading Duration - The difference between Regression Path Duration and Gaze Duration.

Longer time spent rereading prequestions material